

MTGeophysics.jl: A software repository for magnetotelluric research and applications

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Software

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Summary

MTGeophysics.jl is a Julia package for magnetotelluric (MT) geophysical modelling, inversion, and interactive visualization. The magnetotelluric method uses natural electromagnetic signals to image the electrical resistivity structure of the Earth's subsurface, with applications ranging from mineral exploration to studies of tectonic structure. MTGeophysics.jl provides 1D and 2D forward solvers, stochastic inversion based on Very Fast Simulated Annealing (VFSA) (Sen & Stoffa, 2013) in 2D and 3D, I/O routines for the widely used ModEM 3D inversion code (Kelbert et al., 2014), and interactive 3D model viewers built on GLMakie. Because many different subsurface configurations can explain observed MT responses, the VFSA workflow generates an ensemble of plausible models rather than a single deterministic solution, enabling quantitative uncertainty assessment. The package is designed to be accessible to researchers who need an integrated, scriptable environment for MT data processing and interpretation within the Julia ecosystem.

Statement of need

Magnetotelluric surveys produce large volumes of multi-frequency, multi-station electromagnetic data that must be processed through forward modelling and inversion to obtain subsurface resistivity images. Existing MT software is predominantly written in Fortran, MATLAB, or Python: ModEM (Egbert & Kelbert, 2012; Kelbert et al., 2014) is a widely used 3D inversion code written in Fortran; MARE2DEM (Key, 2016) handles 2D/3D marine electromagnetic modelling in Fortran; MTpy (Kirkby et al., 2019) is a Python toolbox for MT data processing and visualization; and numerous MATLAB toolboxes exist for 1D and 2D analysis. However, these tools are often disconnected, with separate programs for forward modelling, inversion, file conversion, and visualization, requiring researchers to maintain ad-hoc scripts in multiple languages to bridge them.

MTGeophysics.jl addresses this fragmentation by providing a single Julia package that spans the entire MT workflow, from data collection and quality control, through forward modelling and inversion, to a final preferred subsurface model with quantified uncertainty. The longer-term vision is to supply all the tools a researcher needs to carry out magnetotelluric studies end-to-end, and to serve as a foundation on which new methods can be built. Because Julia's type system and multiple dispatch make it straightforward to compose packages, MTGeophysics.jl is designed as a core component of a broader JuliaGeophysics ecosystem. Its forward solvers and standardised data structures are intended to support training of neural operators and neural surrogates that learn to approximate the MT forward map, enabling scientific machine learning workflows such as physics-informed neural networks and operator learning for rapid approximate inversion. Julia's composability with automatic differentiation and scientific machine learning libraries makes MTGeophysics.jl a natural building block for bridging classical geophysical inversion with modern data-driven approaches.

The package targets MT researchers and students who want a unified, open-source toolkit that leverages Julia's (Bezanson et al., 2017) strengths in numerical computing, automatic differentiation, composability, and interactive graphics. It reads and writes standard ModEM file formats, enabling direct interoperability with the Fortran ModEM code used by many research groups worldwide.

State of the field

Several open-source tools address parts of the MT workflow. ModEM (Egbert & Kelbert, 2012; Kelbert et al., 2014) is the community standard for 3D MT inversion but provides no built-in visualization or 1D/2D forward capabilities. MARE2DEM (Key, 2016) focuses on 2D and 3D marine electromagnetic forward and inverse modelling but is Fortran-based and limited to controlled-source methods. MTpy (Kirkby et al., 2019) provides comprehensive Python utilities for MT data handling, processing, and visualization but does not include forward solvers or stochastic inversion. pyGIMLi (Rücker et al., 2017) and SimPEG (Cockett et al., 2015) are Python frameworks for general geophysical inversion that include MT modules, but their MT-specific functionality is embedded within much larger codebases. None of these packages offer integrated stochastic inversion with ensemble uncertainty quantification in both 2D and 3D.

MTGeophysics.jl contributes a tightly integrated Julia-native package that covers 1D and 2D MT forward modelling, 2D and 3D VFSA inversion with ensemble uncertainty quantification, and interactive 3D model viewing with GIS overlay support. This combination is not available in any single existing package. The choice of Julia provides performance comparable to compiled languages while maintaining the interactivity and rapid prototyping advantages of interpreted environments.

Software design

MTGeophysics.jl is structured as a single Julia module with clearly separated functional layers. The core layer handles ModEM 3D data and model I/O, supporting the standard ModEM file formats for models (WinGLink/WS format) and impedance data. The 1D module implements both analytical recursive impedance solutions and finite-difference solvers on geometrically graded meshes. The 2D module constructs tensor meshes, assembles sparse finite-difference operators for the TE and TM mode Maxwell equations, and solves them using direct sparse factorization via LinearSolve.jl.

The VFSA inversion module (Sen & Stoffa, 2013) uses radial basis function (RBF) parameterization to map a reduced set of control points to the full model grid, enabling efficient stochastic search in a lower-dimensional space. In 2D the package includes its own finite-difference forward engine; in 3D the inversion wraps the ModEM forward solver (Kelbert et al., 2014), ensuring compatibility with established workflows. Multiple independent Markov chains run in parallel, and ensemble statistics (mean, median, standard deviation) are computed across chains to provide uncertainty estimates. Instead of producing a single deterministic model, the workflow generates a set of plausible models that explain the data comparably well, allowing geologists to interpret results while being explicitly aware of the inherent non-uniqueness.

Interactive visualization is handled through GLMakie, providing GPU-accelerated 3D slice viewers (XY, XZ, YZ, and combined) with slider controls, coordinate reprojection via Proj.jl, and optional shapefile overlays for geological maps, faults, and survey boundaries. The package also supports exporting depth slices and cross-sections as georeferenced shapefiles suitable for integration into GIS platforms, closing the loop between geophysical modelling and geological mapping. The visualization layer is conditionally loaded, ensuring the core package functions without OpenGL dependencies.

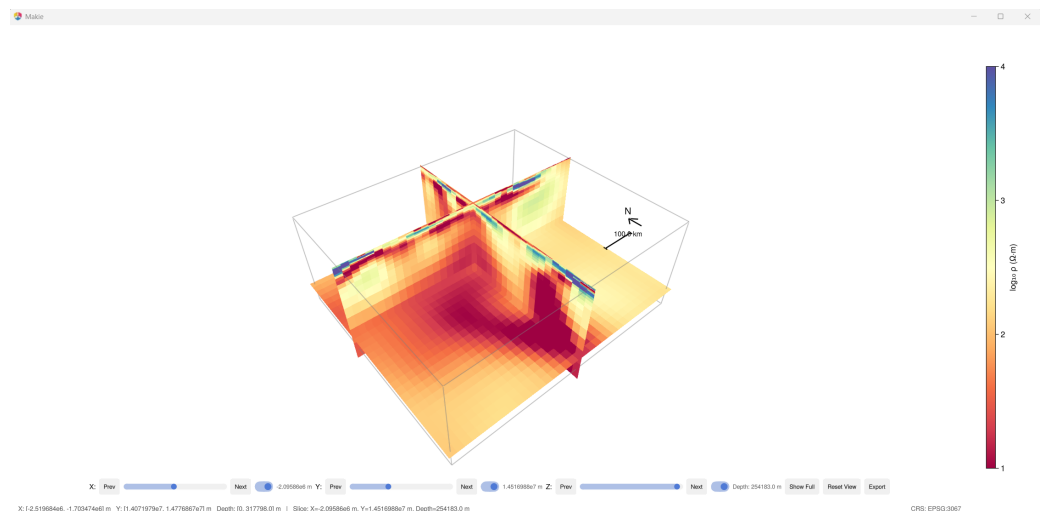


Figure 1: Interactive 3D model viewer showing combined XY, XZ, and YZ resistivity slices with slider controls and coordinate reprojection.

Publication-quality static plots use CairoMakie for data maps, response curves, model cross-sections, and convergence diagnostics. All file formats use plain text, ensuring reproducibility and version-control friendliness.

Research impact statement

MTGeophysics.jl has been developed to support ongoing magnetotelluric research at the Geological Survey of Finland (GTK). The 2D VFSA inversion module has been validated against the COMMEMI benchmarks (Zhdanov et al., 1997), reproducing known synthetic models. The 3D VFSA inversion has been demonstrated at regional scale using USArray MT data from the Cascadia subduction zone (Patro & Egbert, 2008), a dataset extensively studied with deterministic 3D inversion, providing a direct benchmark for the stochastic results. The package's ModEM I/O and 3D visualization capabilities are actively used for interpreting crustal-scale MT surveys in Finland. The repository includes complete reproducible examples with synthetic datasets, benchmark scripts, and documentation, making it ready for adoption by the broader MT research community.

AI usage disclosure

GitHub Copilot was used as a coding assistant during development of this software and in drafting this paper. All AI-generated code and text were reviewed, tested, and verified by the author for correctness.

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